

OPENLEAVE · ARCHITECTURE

The edges, and the core.

How OpenLeave is built and why — a deterministic, citation-backed rules engine at the center, with language models kept to the edges, where they draft and converse but never decide the law.

AUGUST 2026 · the design, in twelve panels

§ 1

THE THESIS

*Every decision that follows falls out
of one commitment.*

LLM at the edges, *engine at the core*.

The LLM belongs at the edges; a verified rules engine belongs at the core.

the survey's organizing thesis, made concrete

Language models draft amendments and hold the conversation. They never decide the law. Taking that seriously forces four properties:

§ The core is **deterministic** — the same facts and date always yield the same answer.

§ Every conclusion **carries its citation** — nothing is asserted without a source.

§ Genuine discretion is **surfaced, not guessed** — open questions come back flagged.

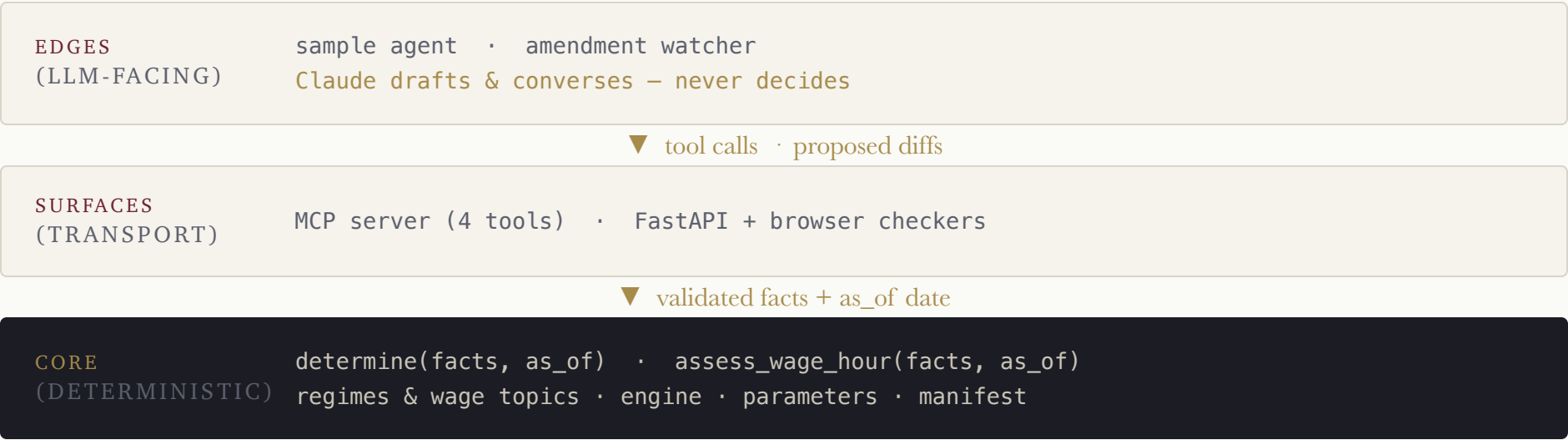
§ The engine **reports the limits of its own coverage** instead of bluffing.

§ 2

STRUCTURE

The core knows nothing of the two tiers above it — everything above is a thin adapter.

Three tiers.



The core is a pure Python library you can call directly; the MCP server, the API, and the agent are thin adapters that transport questions in and cited answers out.

§ 3

PARAMETERS

One mechanism buys "the law as of any date" — and it is what the amendment pipeline maintains.

`openleave/parameters.json`

Numbers stay *data*, on a timeline.

```
# parameters.json — each key is an effective-dated [date, value] series
"minwage.CA": [{"2025-01-01", 16.50}, {"2026-01-01", 16.90}]

parameters.get("minwage.CA", as_of=date(2025, 6, 1)) -> 16.50
parameters.get("minwage.CA", as_of=date(2026, 6, 1)) -> 16.90
```

A wage that clears California's 2025 floor *fails* its 2026 floor. Reconstructing a past determination for an audit is just passing an earlier `as_of`. SAWW bumps, benefit caps, program launch dates, and the vacated-then-restored federal exempt threshold are all just points on a timeline.

§ 4

VOCABULARY

*A leave finding and a wage finding
are the same kind of object.*

`openleave/engine.py`

Everything is a *cited Finding*.

```
Finding(  
    key="serious_health_condition",  
    description="The condition is a 'serious health condition'",  
    met=None,                                # -> a human must decide (never guessed)  
    citation=Citation("29 C.F.R. § 825.113"),  
)
```

`met=None` is load-bearing: it means the question is open-textured and a human must decide it. `resolve_eligibility` propagates that `None` through — discretion is never silently collapsed to a boolean. The same rule flags an exemption's duties test.

§ 5

INVARIANTS

The heart of the design: each rule exists to prevent a specific way a rules-oracle lies.

Built to *refuse* the easy lie.

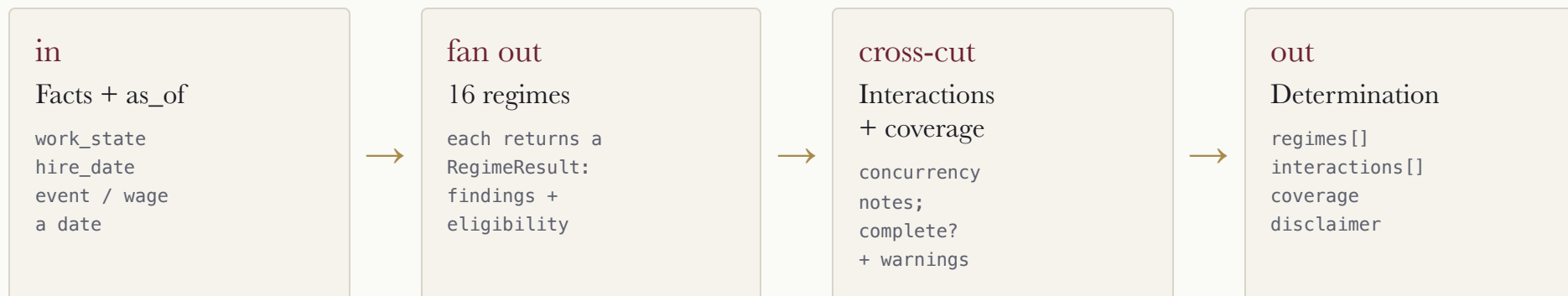
INVARIANT	FAILURE IT PREVENTS
Every conclusion cites its source	Unverifiable or hallucinated law
Discretion is flagged, never compiled (met = null)	Auto-answering open-textured questions — "laundered interpretation"
Coverage is reported, never assumed	Silent under-coverage — a partial answer read as complete
Outside the encoded range ≠ no entitlement	Reading "we lack the historical rate" as "no benefit"
Enacted ≠ in force (e.g. MD FAMILI, 2028)	Fabricating a benefit that cannot be claimed yet
Law as of any date	Answering a past event with today's numbers

§ 6

FLOW

One determination — fanned out to every regime, then reassembled.

Facts in, *cited answer* out.



Each assessor is a pure function of (facts, as_of) plus the read-only parameter store; nothing mutates shared state. `assess_wage_hour` has the identical shape — fan out to topic assessors, attach locality-aware coverage, return a dict.

§ 7

REUSE

Wage & hour was added over the leave engine's substrate, not beside a copy of it.

Wage & hour is a *sibling*, not a rewrite.

LEAVE

determine(facts, as_of)
regimes/ — 16 modules

Facts are **event-shaped** (a LeaveEvent)

WAGE & HOUR

assess_wage_hour(facts, as_of)
wagehour/ — topic assessors + localities
WageFacts are **ongoing-state** (no single event)

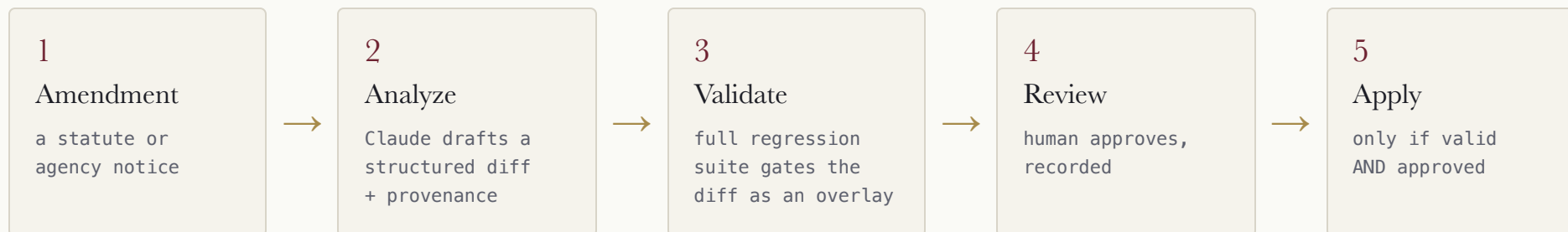
SHARED SUBSTRATE Both reuse the same Finding / Citation types, the effective-dated parameter store, the coverage-reporting reflex, and the verification manifest. Only the fact shape, the entry point, and the locality dimension are new — the template for a third domain (breaks, WARN-Act) to follow.

§ 8

CURRENCY

*Encoding the law once is easy;
staying current is the product. The
LLM proposes — a human and the
tests decide.*

Staying current, *safely*.



THE GUARANTEES *Parameter updates are machine-appliable; structural changes are flagged `requires_human_encoding` and never auto-applied. A diff that rewrites an in-force historical value breaks pinned determinations and is rejected. Nothing ships without passing tests and a recorded sign-off.*

§ 9

TRUST

*The one thing standing between an
impressive prototype and reliance.*

`openleave/references.json`

The trust boundary, *drift-guarded*.

Every statutory value here is web-researched and **unverified by counsel** — the load-bearing caveat. The manifest is the worksheet for closing it.

- § Maps **every parameter** to a plain-English meaning, its governing statute, and the agency page to check it against — with per-jurisdiction sign-off fields.
- § A test asserts it documents **exactly** the encoded parameter set — so a rate cannot be added without a citation, and the build fails if one is.
- § No jurisdiction can be marked verified without a **named reviewer and date**.
- § The MCP lookup tool returns each value's source and verification status, so a consumer always knows a figure is still **pending review**.

§ 10

EDGES & EXTENSION

*The seams are guarded so you
cannot add law without documenting
its source.*

Thin edges, *guarded seams*.

THE SURFACES

All three — **MCP server** (4 tools), **REST API + browser checkers**, and the **sample agent** — call the same two core functions and add nothing to the reasoning. The agent is the only place a model answers a user, and even there it is forbidden to answer substantive questions except through the tools.

WHERE TO EXTEND

Add a leave **regime**, a wage **topic**, or a **jurisdiction** — the recurring pattern is that the manifest drift-guard and the tests **force you to document its source**. That is the architecture enforcing its own central promise: no law without a citation.

DESIGN IN ONE BREATH,

A deterministic, cited core;
the language, at the edges.

OPENLEAVE · employment law, as code

github.com/geneweng/cc_rules_as_code · see ARCHITECTURE.md for the full document